

Do the arXiv papers support a dangerous LHC black-hole scenario?

Processed arXiv evidence: claims, equations and physical transfer

Abstract

The dangerous LHC black-hole scenario fails at the mechanism level. A catastrophe requires one continuous physical branch: production in a collision, survival against evaporation, stopping or capture in matter, net positive mass growth, and evasion of astronomical survival bounds. The claim graph contains 869 nodes and 377 source-to-claim links. It places the source set: mostly astrophysical black-hole material, a smaller risk surface, and very few explicit safety statements. The equation graph tests the physical branch. From 1,408 processed equation windows, 204 usable equation nodes remain; 72 touch the LHC black-hole question and 35 remain as branch receipts. Those receipts populate astronomical growth, accretion, compact-object and lifetime constraints, plus one collider selection hook. The collider-growth branch is broken at the required links from production to survival, from survival to capture, and from capture to fast positive growth. The scientific result is a transfer calculation: astrophysical rate and bound equations constrain any microscopic collider-product scenario, and the retained equations stop before the missing bridge.

The public question is usually phrased as whether the LHC could make a dangerous black hole. That wording hides the physics. A dangerous outcome would require a sequence of events, not a single production claim: a microscopic object must be created in a parton collision, survive evaporation, slow down or be captured by matter, accrete faster than it loses mass, and remain compatible with the survival of astronomical bodies exposed to much higher natural collision histories. The analysis below turns that sequence into an equation-level constructor. The claim graph shows what the literature says; the mechanism graph tests which steps are actually filled by formulas and where the physical branch stops.

1 Mechanism verdict

A dangerous LHC branch must connect a collider event to a long-lived object, then to stopping or capture in matter, then to positive mass growth on a relevant time scale. The retained equations fill adjacent astrophysical branches: accretion rates, luminosity limits, compact-object masses, merger parameters and survival constraints. The collider side contributes a threshold or event-selection hook. The later growth chain remains absent.

The physical question is therefore concrete. Let s denote the squared proton–proton centre-of-mass energy at the collider, x_1 and x_2 the parton momentum fractions, and $\hat{s} = x_1 x_2 s$ the parton-level squared centre-of-mass energy. Let M_D denote the fundamental gravitational scale in the low-scale-gravity model, $M_{\min}$ the minimum black-hole mass assumed by that model, M the mass of the candidate microscopic black hole, τ_{evap} its evaporation lifetime, τ_{capture} the time required for stopping or capture in matter, ρ the density of the material it traverses, $\sigma_{\text{cap}}(M, v)$ the effective capture or accretion cross-section, v the speed relative to the material, $P_{\text{evap}}(M)$ the evaporation power, and c the speed of light. In this notation the minimum danger chain is

$$\sqrt{\hat{s}} = \sqrt{x_1 x_2 s} > M_{\min} \sim \text{few } M_D, \quad \tau_{\text{evap}} > \tau_{\text{capture}}, \quad \dot{M}_{\text{net}} = \rho \sigma_{\text{cap}}(M, v) v - \frac{P_{\text{evap}}(M)}{c^2} > 0. \quad (1)$$

Equation (1) states the three required steps: the collision can access the new-gravity channel, the object survives long enough to be captured, and the net mass-growth rate $\dot{M}_{\text{net}}$ is positive. The chain also requires a growth time short enough to matter and no contradiction with compact-object survival. The retained equations leave this chain open.

2 Physical constructor

In this report, a constructor is an ordered physical branch with acceptance tests at each step. It is closer to a derivation recipe than to a topic label: each step must supply a quantity that the next step can use. For the LHC black-hole question the constructor has six slots. The production slot asks whether a parton collision can enter the low-scale-gravity channel. The survival slot asks whether the object lives long enough to interact with matter. The capture slot asks whether it can stop or remain bound inside matter. The growth slot asks whether intake exceeds evaporation. The timescale slot asks whether the integrated growth is fast enough to matter. The astronomical-bound slot asks whether the same mechanism would contradict the survival of dense astronomical objects.

Each slot is represented by a minimal equation condition:

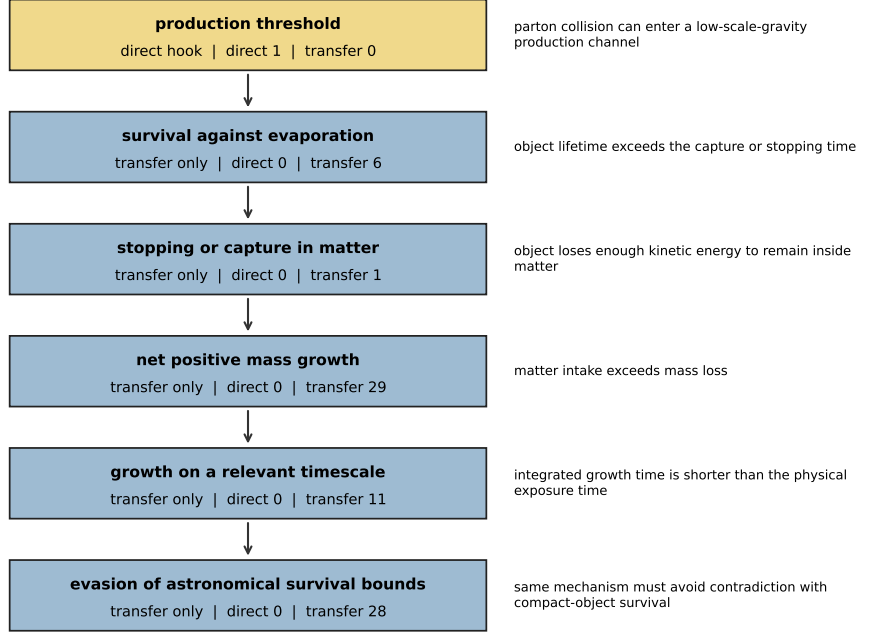
$$\begin{array}{ll}
 \text{production:} & \sqrt{\hat{s}} > M_{\min}, \\
 \text{survival:} & \tau_{\text{evap}} > \tau_{\text{capture}}, \\
 \text{capture:} & L_{\text{stop}} < L_{\text{body}}, \\
 \text{net growth:} & \dot{M}_{\text{net}} > 0, \\
 \text{growth time:} & t_{\text{grow}} < t_{\text{exposure}}, \\
 \text{astronomical closure:} & N_{\text{CR}} P_{\text{cap}} P_{\text{grow}} \ll 1.
 \end{array} \tag{2}$$

Here L_{stop} is the stopping length, L_{body} the material length scale, t_{grow} the time required for dangerous growth, t_{exposure} the relevant exposure time, N_{CR} the number of cosmic-ray production trials in the astronomical comparison, P_{cap} the probability of capture and P_{grow} the probability of subsequent growth. These conditions are intentionally minimal. They specify what must be shown before the scenario can become a closed physical branch.

The constructor then classifies equation receipts. A direct receipt is a formula written in collider variables for the slot it fills. A transfer receipt is a formula from an adjacent astrophysical setting that carries the same physical role, such as accretion rate, lifetime, capture scale or compact-object bound. Transfer receipts are valuable because they show the relevant physics and the variables that must be translated. Branch closure, however, requires direct collider receipts for the downstream slots and composability between slots: production must feed survival, survival must feed capture and capture must feed positive growth. In the retained corpus, production has a direct hook; the downstream slots are transfer-only.

The constructor layer is built from the retained equation fingerprints, not from paper abstracts. It expands the six-slot summary into 1,408 fingerprinted equation windows from 241 sources. Of these, 986 instantiate at least one constructor slot and 412 are relevant to the LHC black-hole question. The source-local graph contains 241 constructor chains. This is the layer that justifies the mechanism verdict: it is a slot-by-slot assembly of measured equation roles, not a count of papers.

Required physical branch: every slot must be filled by collider-relevant equations



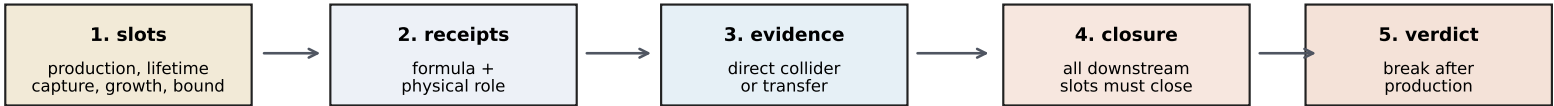
Result: the corpus contains a production hook and astrophysical transfer equations, but no direct collider chain for lifetime, capture and growth.

Figure 1: Physical constructor for the danger branch. The first slot has a collider-side production hook. The downstream slots are filled only by transfer equations from astrophysical black-hole, accretion or compact-object papers. Thus the continuous branch from LHC production to survival, capture and fast growth is not present in the retained equations.

Table 1: Evidence status of each required physical slot.

Physical slot	Status	Direct	Transfer	Required condition
production threshold	direct hook	1	0	parton collision can enter a low-scale-gravity production channel
survival against evaporation	transfer only	0	6	object lifetime exceeds the capture or stopping time
stopping or capture in matter	transfer only	0	1	object loses enough kinetic energy to remain inside matter
net positive mass growth	transfer only	0	29	matter intake exceeds mass loss
growth on a relevant timescale	transfer only	0	11	integrated growth time is shorter than the physical exposure time
evasion of astronomical survival bounds	transfer only	0	28	same mechanism must avoid contradiction with compact-object survival

Constructor operation: a danger branch is a sequence of required physical slots, not a count of papers.



physical slot	slot equation template	direct collider	transfer analogue	slot status
production threshold	$\sqrt{s} > M_{\min}$	1		0 direct hook
survival lifetime	$\tau_{\text{evap}} > \tau_{\text{capture}}$	0		6 transfer only
stopping / capture	$L_{\text{stop}} < L_{\text{body}}$	0		1 transfer only
net mass growth	$\dot{M}_{\text{net}} > 0$	0		29 transfer only
growth timescale	$t_{\text{grow}} < t_{\text{exposure}}$	0		11 transfer only
astronomical bound	$N_{\text{CR}} P_{\text{cap}} P_{\text{grow}} \ll 1$	0		28 transfer only

no downstream
direct collider
closure

Direct evidence closes a collider slot. Transfer evidence marks a relevant physical role that still has to be rewritten and tested in collider variables.

Figure 2: How the constructor works. The upper row shows the operation: define the physical slots, match formula receipts to those slots, separate direct collider evidence from transfer analogues and then test whether the branch closes continuously. The lower matrix shows the actual fill state for this run. Production has one direct hook; every downstream safety-critical slot is transfer-only, so the danger branch does not close.

3 Discourse graph and mechanism graph

The same source set gives two different graphs. The first graph is a discourse graph: 492 paper nodes, 377 claim nodes and 377 source-to-claim links. It records attribution and coverage: most extracted statements in this run are astrophysical black-hole statements, with a smaller risk surface and very few explicit safety statements.

The second graph is an equation mechanism graph. It starts from formula windows, keeps the windows that contain usable mathematical structure, assigns operator/substrate constructor roles, and asks whether those formulas fill the physical slots in Fig. 1. Its sparse attention is not over words. It is over physical roles: threshold, lifetime, capture, growth and astronomical-bound closure. This graph contains 35 evidence-grade receipts for the case layer. The decisive comparison is that the discourse graph can count claim families, whereas the mechanism graph can test branch closure. For the required downstream slots, the constructor finds 0 direct collider closures and 5 transfer-only slots.

Sparse-attention proof: discourse finds claims; the mechanism layer tests branch closure

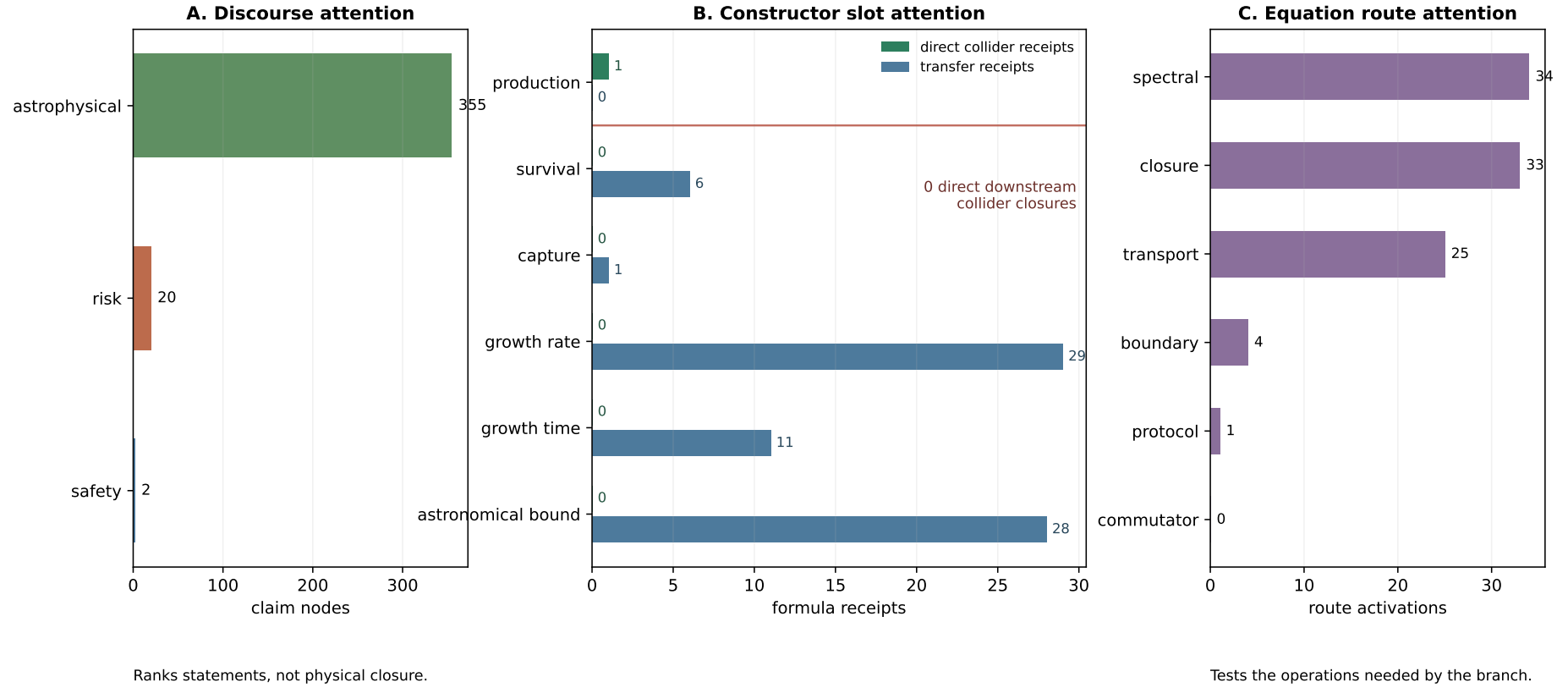


Figure 3: Discourse graph versus mechanism graph on the same source set. Panel A shows claim-family attention: it ranks statements. Panel B shows the physical constructor: it tests whether required equation slots are filled by direct collider receipts or only by transfer analogues. Panel C shows route attention inside the retained equation receipts. The claim graph locates statements; the mechanism graph identifies the missing physical bridge from production to survival, capture and growth.

The comparison changes the question from literature coverage to physical closure. Source and claim counts show where statements occur. The constructor asks a stricter question: which formulas occupy the required physical roles, and do those roles connect into a continuous collider branch? Clean formula extraction, operator/substrate constructor roles, sparse route attention and slot-by-slot assembly give that answer directly.

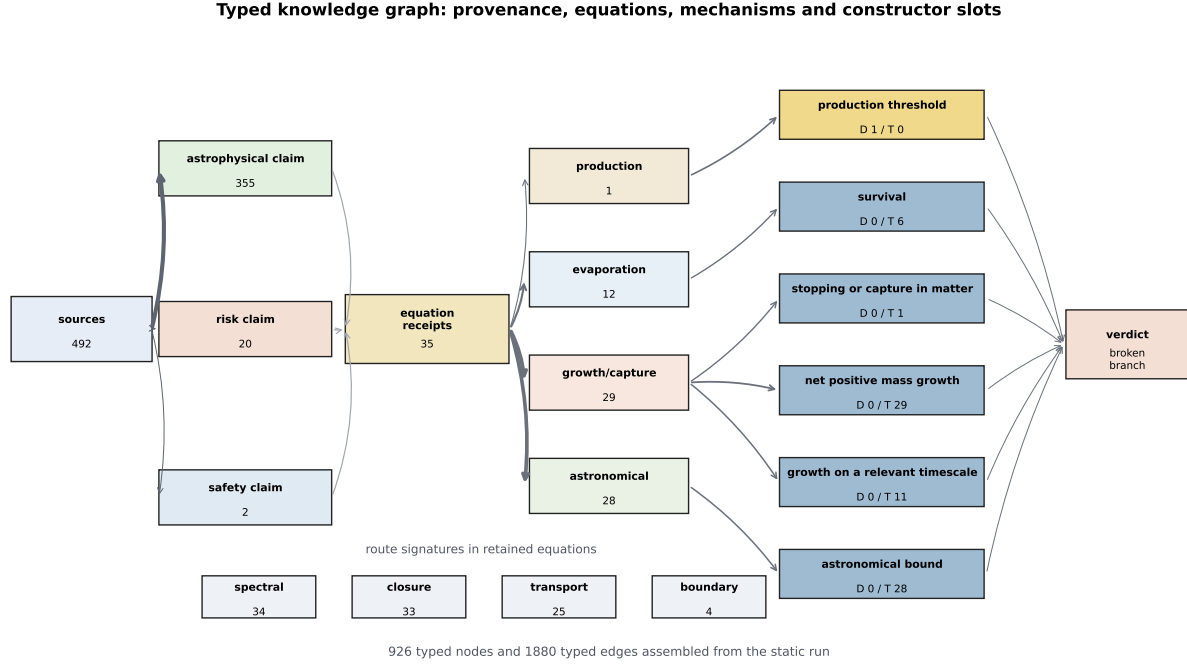


Figure 4: Typed knowledge graph assembled from the static run. It contains 926 typed nodes and 1,880 typed edges. The left side is provenance; the middle is the retained equation layer; the right side is the constructor layer that decides whether the danger branch closes.

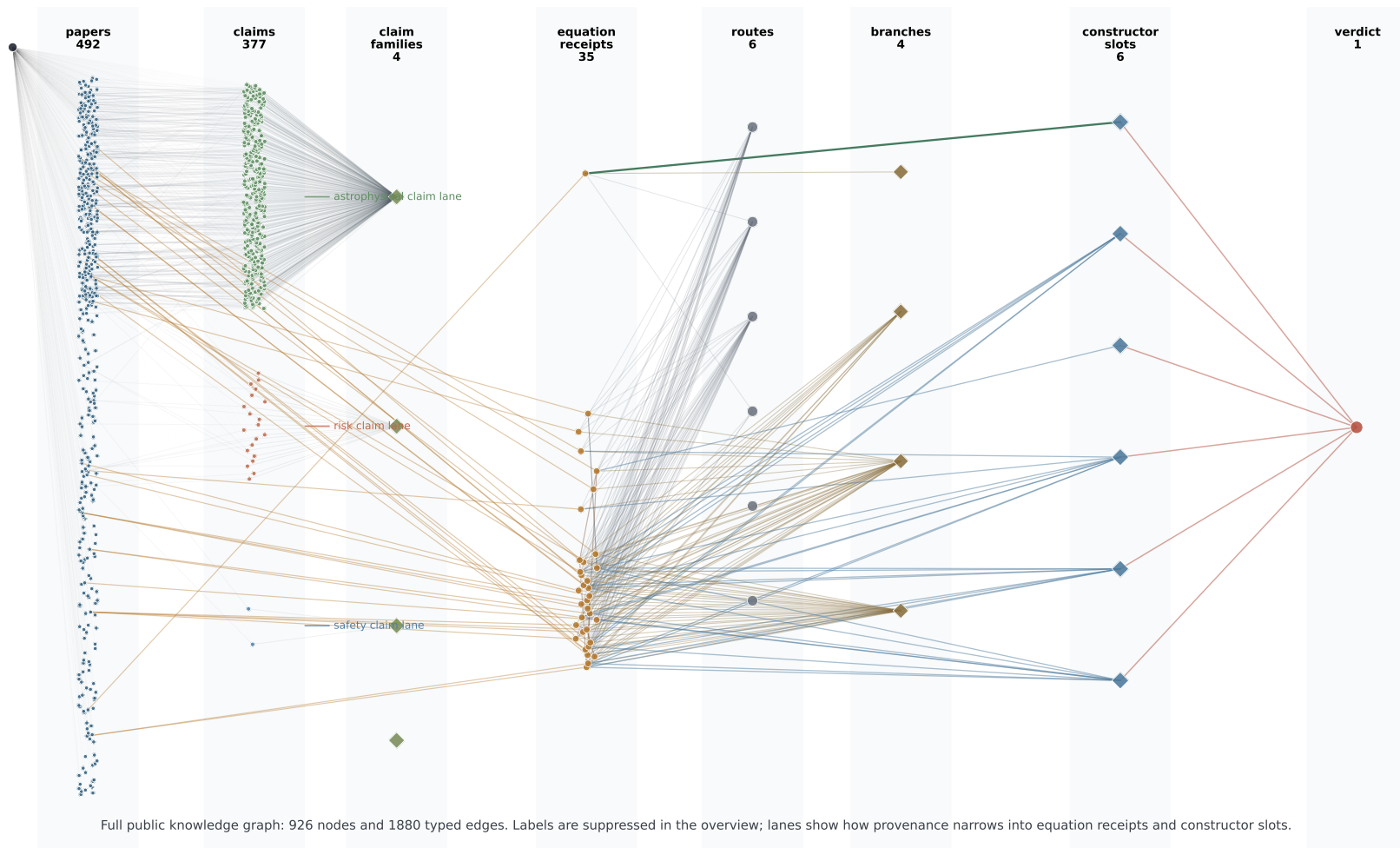


Figure 5: Full public knowledge graph. Each point is one node in the 926-node graph. The dense paper and claim layers remain on the left; sparse equation receipts, route labels, physical branches and constructor slots appear on the right. Labels are omitted in this overview because the graph is topology-first; source names and formulas are given in the receipt tables and machine-readable JSON. The coordinates are layout coordinates only: horizontal position separates node layers, while vertical position groups claim families and physical branches and spreads overlapping nodes.

4 How formation is supposed to work

The LHC formation hypothesis used in these papers belongs to low-scale gravity models. The fundamental gravitational scale is lowered to the TeV range, for example by large extra dimensions or many particle species. The collider energy $\sqrt{s}$ is shared by two partons carrying momentum fractions x_1 and x_2 . The parton-level squared centre-of-mass energy is $\hat{s}$, defined by

$$\sqrt{\hat{s}} = \sqrt{x_1 x_2 s}. \quad (3)$$

Formation is then treated as possible only if the parton-level energy exceeds a model-dependent minimum black-hole mass $M_{\min}$. In the collider papers this threshold is usually expressed relative to M_D , the fundamental gravitational scale:

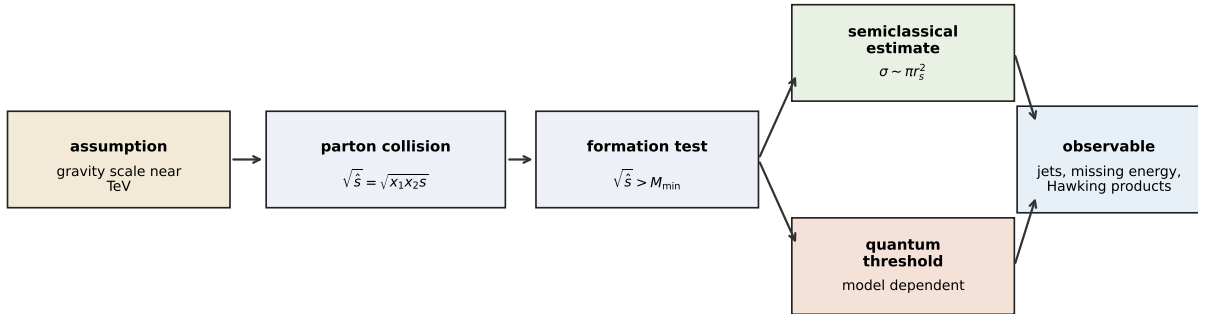
$$\sqrt{\hat{s}} > M_{\min} \sim \text{few } M_D. \quad (4)$$

In the semiclassical picture the production cross-section σ_{form} is estimated from a higher-dimensional horizon radius r_s . Here M_{BH} is the produced black-hole mass and n is the number of extra dimensions:

$$\sigma_{\text{form}} \sim \pi r_s^2(M_{\text{BH}}, M_D, n). \quad (5)$$

Close to threshold, Eq. (5) becomes a quantum-gravity assumption rather than a clean classical black-hole calculation. Formation papers can predict detector signatures; a danger argument still requires survival, capture and positive growth.

LHC microscopic black-hole formation hypothesis



Formation is the first step. Danger also requires survival, stopping, capture and positive growth in matter.

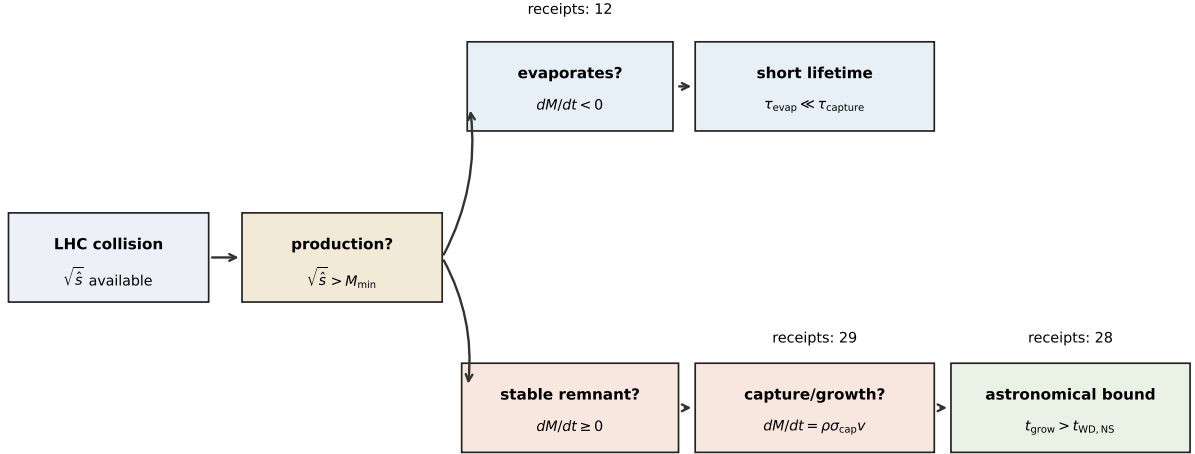
Figure 6: Formation-side mechanism. The LHC black-hole hypothesis begins with a low gravitational scale, then applies a parton-level threshold. Above threshold, some papers use a geometric cross-section estimate; near threshold the object is model-dependent quantum gravity. The later growth-in-matter branch is a separate physical requirement.

Table 2: Formation-side sources and what they actually support.

Source	Formation-side role	Formula or scale	Meaning
hep-ph0111052	large-extra-dimension production and high transverse-momentum jet suppression	$\sigma(pp \rightarrow \text{jet} + X)$	production/signature model; no growth-in-matter term
hep-ph0605062	TeV-scale large-extra-dimension black holes	$M_f = 1 \text{ TeV}; \sqrt{s} = 14 \text{ TeV}; M > M_f$	requires a low gravity scale; rate is model-dependent
hep-ph0602129	Higgs production from black-hole evaporation	$T_{\text{BH}} \sim 1 \text{ TeV}$	decay/signature channel; accretion untested
0904.0230	TeV black-hole decay products	$p_T^H > 100 \text{ GeV}; y_{\gamma\gamma} \leq 1; M_P = 1 \text{ TeV}$	detector selections after assumed production
0806.3801	many-species microscopic black holes	$M_{\text{Planck}}/\sqrt{N}$	collider threshold changes only in modified gravity
0710.4344	species bound and lowered gravitational cutoff	$\Lambda_G \approx M_{\text{Planck}}/\sqrt{N}$	formation tied to a lowered cutoff beyond Standard Model assumptions

The formation-side result is specific. The selected literature contains papers on possible production and possible decay signatures of TeV-scale black holes. These formulas support search channels: high- p_T suppression, Hawking-like decay products, heavy-particle production and model-dependent threshold behavior. The safety-critical steps remain separate: survival, stopping, capture and sustained accretion. Formation occupies the first box in Fig. 7.

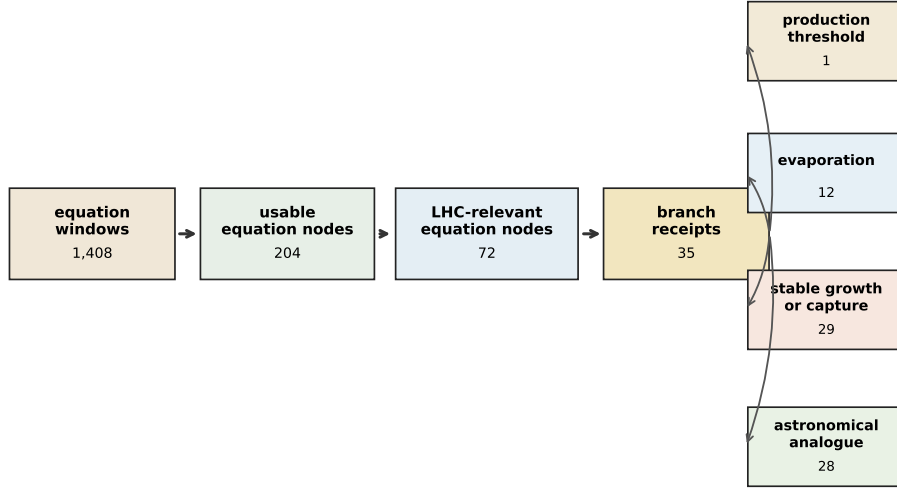
Branched physical tree for the LHC black-hole catastrophe claim



The catastrophe branch requires production, survival, capture, rapid growth, and failure of the astronomical bound.

Figure 7: Physical knowledge tree for the LHC black-hole catastrophe claim. A dangerous branch requires all steps: production, survival, capture, fast growth and failure of astronomical bounds. The retained equations populate the adjacent astronomical and growth/capture parts of this tree.

From processed equations to the retained physical branch receipts



Result: equations accumulate around adjacent astrophysical constraints; no direct survival-capture-growth chain is retained for the collider case.

Figure 8: Equation funnel for the LHC black-hole question. The graph starts with 1,408 equation windows and retains 204 usable equation nodes. Of these, 72 are relevant to the LHC black-hole question and 35 remain as branch receipts. Branch counts overlap because one equation can support more than one physical branch.

5 The claim graph

The provenance graph contains 492 paper nodes, 377 claim nodes and 377 source-to-claim links. Its result is simple: the selected source set contains many papers where black holes appear in astrophysical settings, fewer explicit risk statements, and very few explicit safety statements.

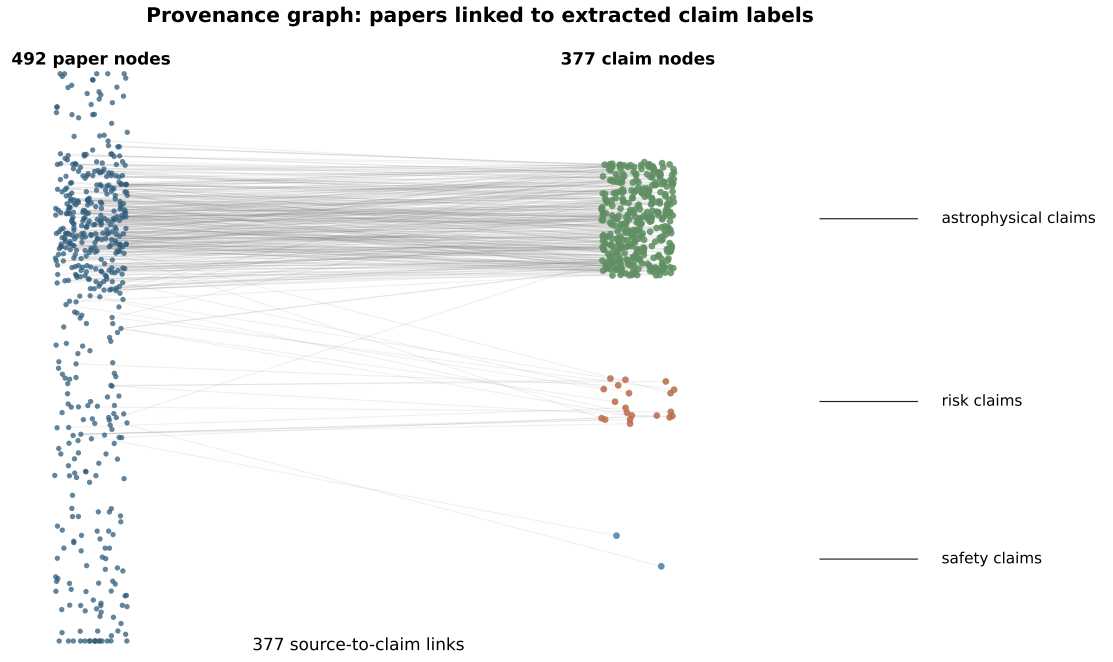
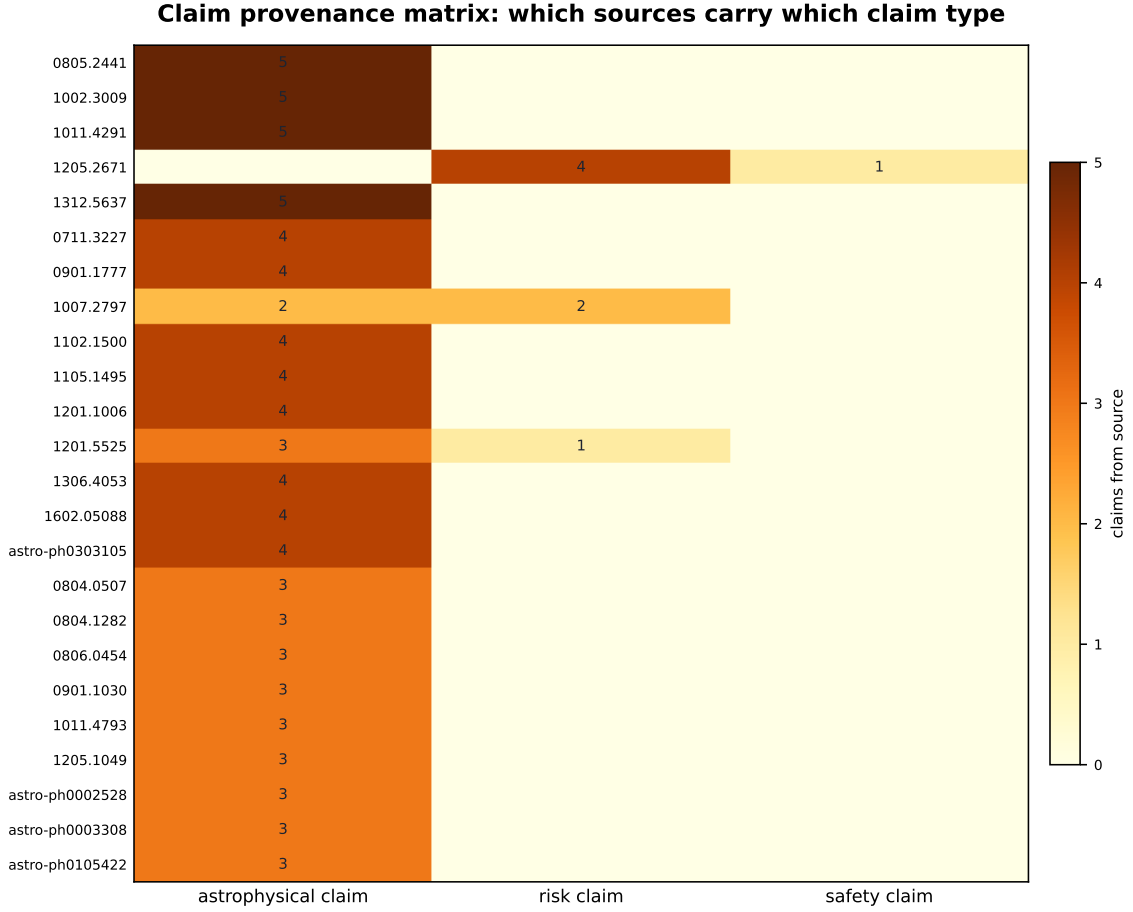


Figure 9: Full claim graph. Paper nodes are on the left; extracted claim nodes are on the right; each line is a source-to-claim link. This graph contains 492 papers, 377 claims and 377 links. The coordinates are graph-layout coordinates only: horizontal position separates papers from claim nodes, while vertical position groups claim families and adds jitter so overlapping nodes can be seen.



Rows show the highest-claim sources. Numbers are extracted source-to-claim links, not aggregated topic labels.

Figure 10: Source-level provenance matrix. Rows are the highest-claim sources; columns are extracted claim families. Cell values are individual source-to-claim links. This view preserves provenance rather than collapsing the literature to one aggregate edge.

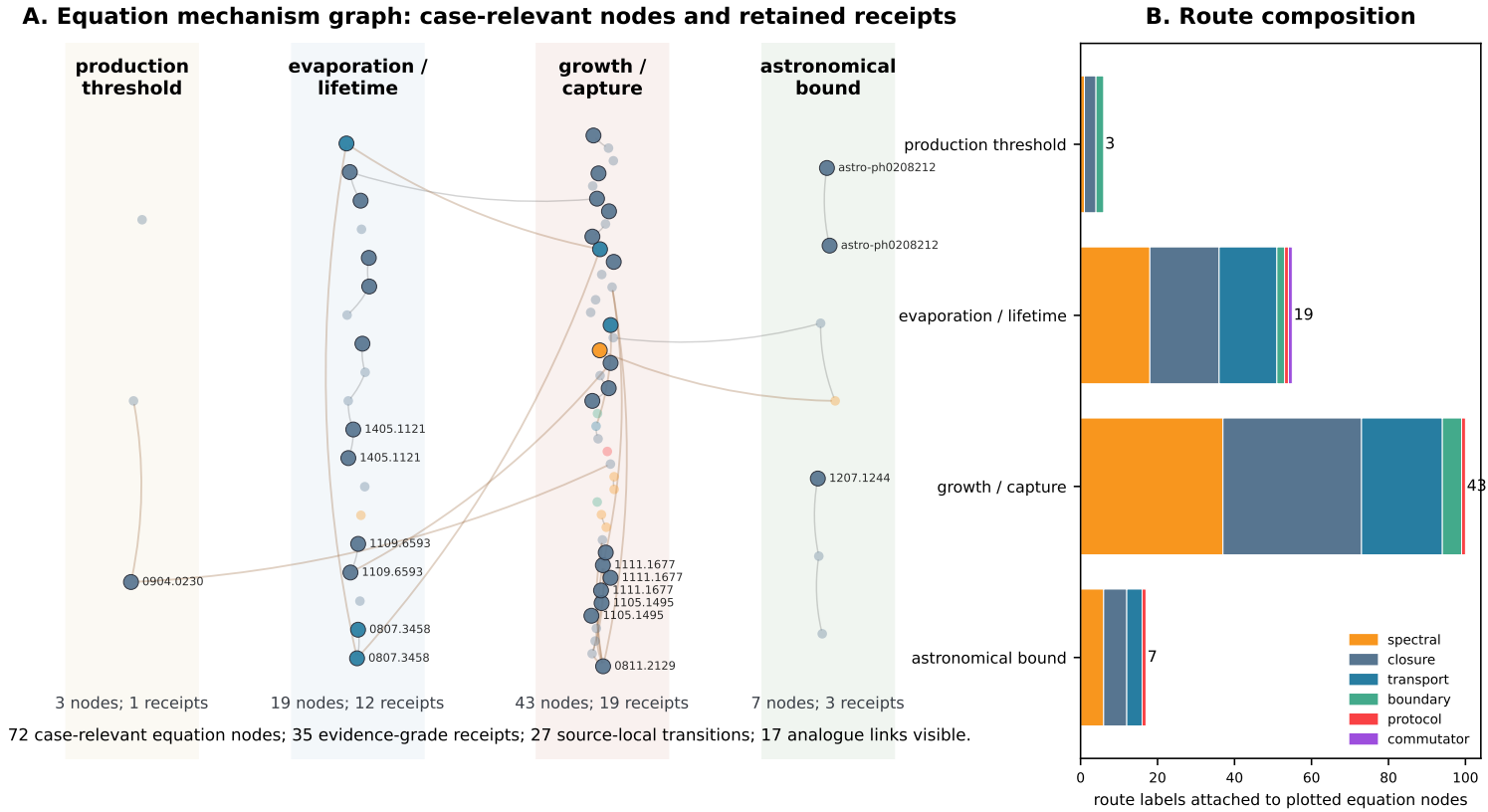
The claim labels are:

Quantity	Count
astrophysical claim	355
risk claim	20
safety claim	2

The claim graph gives the social surface of the literature: where black-hole, risk and safety statements appear. The physical question starts after that. The required route is a mathematical one, from LHC production to survival, capture and growth.

6 The equation graph

The equation graph tests whether formulas assemble into the danger chain. It retains 35 branch receipts. They sit mainly on astronomical analogues and stable growth/capture constraints. The missing step is specific: no retained formula chain directly connects an LHC-produced object to survival, stopping or capture in ordinary matter, and net positive growth on a dangerous timescale.



Interpretation: the graph is dense in spectral/closure/transport roles, but the collider side has only a production hook; survival, capture and growth are supplied as transfer constraints, not as a direct collider derivation.

Figure 11: Retained equation graph. Each point is a formula receipt placed into one physical branch. Panel A shows all case-relevant equation nodes, with the evidence-grade receipts drawn larger and darker. Edges are source-local transitions or cross-source route analogues. Panel B shows which mathematical routes are attached to the plotted nodes. The graph shows that retained formulas are dense in spectral, closure and transport roles, but the collider side has only a production hook. The survival–capture–growth steps are transfer constraints drawn from adjacent astrophysical mechanisms, not a direct collider derivation in the selected corpus.

The retained branches are:

Quantity	Count
stable growth or capture branch	29
astrophysical black hole analogue	28
evaporation branch	12
production threshold branch	1

The retained equations support rate laws, constraints and scale relations for compact objects. The missing physical bridge is the one from collider production to survival, stopping, capture and fast growth inside ordinary matter.

7 Equation receipts

Representative receipts are shown in Table 3. Each receipt is a local mathematical object that places a source inside one branch of the physical tree. The identifiers R1–R7 refer to source-local formula windows; the report-level variables used for the LHC mechanism are defined in Eqs. (1)–(7).

Table 3: Representative equation receipts from the retained branch layer.

Receipt	Branch	Equation receipt	Interpretation
R1 0904.0230	production threshold	$ y_{\gamma\gamma} \leq 1$	Collider-side selection or threshold condition. It is a production hook; the survival and growth steps remain separate.
R2 0807.3458	astronomical analogue; evaporation or lifetime; stable growth/capture	$3 \times 10^{-13} \text{ M}_\odot \text{ yr}^{-1}$ $\lesssim \langle \dot{M}_{\text{long}} \rangle$ $\lesssim 1 \times 10^{-10} \text{ M}_\odot \text{ yr}^{-1}$	Mass-change or lifetime evidence. It belongs to the branch that can close the risk by evaporation or slow evolution.
R3 0807.3458	astronomical analogue; evaporation or lifetime; stable growth/capture	$4 \times 10^{-14} \text{ M}_\odot \text{ yr}^{-1}$ $\lesssim \langle \dot{M}_{\text{long}} \rangle$ $\lesssim 2 \times 10^{-11} \text{ M}_\odot \text{ yr}^{-1}$	Mass-change or lifetime evidence. It belongs to the branch that can close the risk by evaporation or slow evolution.
R4 astro-ph0212297	astronomical analogue; evaporation or lifetime; stable growth/capture	$B_c \simeq 10^{16} \text{ G} (7 \text{ M}_\odot / M_H)$ $(6 M_H / R)^2$ $(M_T / 0.03 M_H)^{1/2}$	Mass-change or lifetime evidence. It belongs to the branch that can close the risk by evaporation or slow evolution.
R5 astro-ph0109539	stable growth/capture	$\langle M \rangle \sim 9 \text{ M}_\odot$	Growth, capture or compact-object scale evidence. It tests the stable-remnant branch.
R6 1207.1244	astronomical analogue	$M_1 = 0.9^{+4.6}_{-0.3} \text{ M}_\odot$	Astronomical analogue evidence. It constrains whether a similar growth mechanism is compatible with observed compact objects.
R7 1307.7685	astronomical analogue; stable growth/capture	$\delta M_{BH} < 5 \times 10^{-4} M_{BH}$	Growth, capture or compact-object scale evidence. It tests the stable-remnant branch.

8 What the equations imply

The selected equations support a transfer calculation. Astrophysical systems provide constraints on growth, accretion, luminosity and compact-object survival. Let $\dot{M}_{\text{astro}}$ denote an astrophysical mass-growth rate and $\dot{M}_{\text{LHC}}$ the corresponding growth term proposed for a microscopic object moving through matter. The LHC scenario can borrow astrophysical equations only if the same physical role survives the change of scale:

$$\dot{M}_{\text{astro}} \longrightarrow \dot{M}_{\text{LHC}} = \rho \sigma_{\text{cap}}(M, v) v. \quad (6)$$

Let $t_{\text{grow}}^{\text{astro}}$ be the astrophysical growth time inferred from compact-object systems, and let $t_{\text{WD,NS}}$ denote the observed survival time of white dwarfs or neutron stars used as astronomical bounds. A collider translation must replace that astrophysical time-scale by the microscopic capture and evaporation variables already defined above:

$$t_{\text{grow}}^{\text{astro}} > t_{\text{WD,NS}} \longrightarrow \{\tau_{\text{capture}}, \tau_{\text{evap}}, \rho, \sigma_{\text{cap}}(M, v), v\} \quad \text{for a microscopic object in matter.} \quad (7)$$

Equations (6) and (7) are valid only where the same growth or capture mechanism is preserved. Evaporation, failed capture, or slow growth each breaks the dangerous branch.

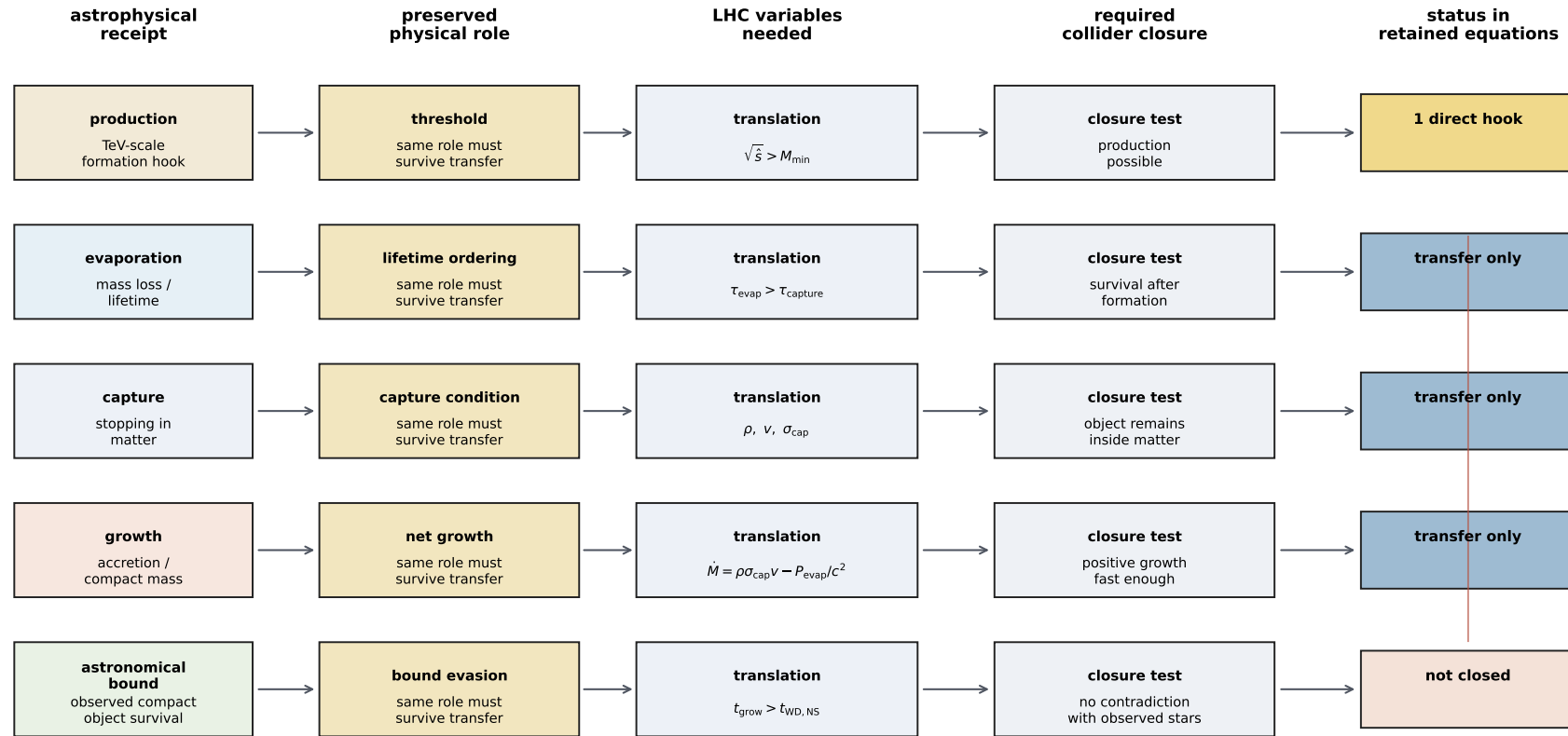
Plain-language result:

- Production alone gives the starting condition. A threshold condition says where the scenario could begin.
- Evaporation closes the danger branch if the object loses mass before it can be captured.
- Stable survival gives only a lifetime condition. A stable object must also be stopped, captured and made to grow.
- Growth requires a medium, a cross-section and time. In equations this is the role of ρ , $\sigma_{\text{cap}}(M, v)$, v and the growth timescale.
- Astronomical bounds matter because long-lived compact objects test whether similar growth mechanisms would already have destroyed observed systems.
- In this corpus the equations mostly support the astronomical-bound side of the argument. The absent formula is not just a missing citation. It is the specific constructor step

$$\text{LHC production} \rightarrow \text{survival} \rightarrow \text{stopping/capture} \rightarrow \dot{M}_{\text{LHC}} > 0$$

with a growth timescale short enough to matter. The retained equations do not provide that source-local collider chain; they provide transfer constraints against it.

Equation transfer: what must survive when astrophysical black-hole equations are moved to an LHC object



astrophysical analogue receipts retained: 28

direct collider-safety closures retained: 0

Result: the literature supplies adjacent mechanisms and bounds, but it does not supply the continuous collider survival-capture-growth chain.

Figure 12: Equation transfer from astrophysics to the LHC case. The retained equations constrain growth, luminosity, compact-object survival and accretion in astrophysical settings. Each row shows the required transfer: an astrophysical receipt must preserve its physical role, be rewritten in LHC variables and pass the corresponding collider closure test. The retained corpus supplies transfer receipts but no direct collider closure beyond the production hook.

9 Alignment with CERN-2003-001 and equation correctness

CERN-2003-001, the report of the LHC Safety Study Group, studied potentially dangerous events in heavy-ion collisions and treated negatively charged strangelets, gravitational black holes and magnetic monopoles [1]. Its gravitational section uses the same branch structure used here. Production is only the first step. A dangerous case also requires stability against thermal decay, accretion faster than decay, and a mechanism that remains compatible with ordinary matter and astrophysical constraints. The agreement is structural: both arguments test whether production, survival, capture, growth and astronomical closure can be connected into one physical branch.

For ordinary four-dimensional gravity, CERN writes the Schwarzschild radius as

$$R_S = 2GM = \frac{M}{M_P^2}, \quad (8)$$

where G is Newton's constant and M_P is the four-dimensional Planck mass. Comparing R_S with the LHC localization scale gives a required mass far beyond accelerator energies. CERN then compares the Hawking decay rate

$$\Gamma_D \simeq T_{\text{BH}}^4 R_S^2 \quad (9)$$

with an upper bound on accretion through matter,

$$\Gamma_A \simeq \pi R_S^2 \rho. \quad (10)$$

Here T_{BH} is the black-hole temperature and ρ is the density of matter. Growth requires $\Gamma_A > \Gamma_D$, which CERN rewrites as a mass bound many orders of magnitude above the LHC. In models with d extra dimensions and fundamental scale M_* , CERN uses

$$T_{\text{BH}} \simeq M_* \left(\frac{M_*}{M} \right)^{1/(1+d)} \quad (11)$$

and again obtains a stability mass far beyond accelerator production, even in the extreme case $M_* = 1 \text{ TeV}$. Stable extremal black holes become a growth mechanism only if ordinary matter carries the conserved charge needed to feed them.

The equation status is as follows. Equation (1) is the constructor form of the danger branch, with the production condition written at parton level through $\hat{s} = x_1 x_2 s$. Equation (8) is the standard Schwarzschild-radius relation in units $c = \hbar = 1$. Equations (9) and (10) are the same scaling comparison used in CERN-2003-001: Hawking loss must be beaten by matter intake before growth can begin. Equation (11) is the higher-dimensional temperature scaling used for the TeV-gravity case. The rate-balance term in Eq. (1) is the CERN decay-versus-accretion comparison written as a sign test for $\dot{M}_{\text{net}}$.

The present analysis adds four elements to the CERN report. First, it separates the literature into a discourse graph, which records who states which kind of claim, and an equation graph, which records which formulas occupy the physical slots of the branch. Second, it keeps direct collider receipts separate from astrophysical transfer receipts, so an accretion equation from a compact-object paper cannot be mistaken for a collider closure. Third, it shows the missing links explicitly: after the production hook, the survival, capture and growth slots are filled only by transfer equations. Fourth, it makes the CERN safety logic inspectable on a corpus: any new paper can be tested by asking which slot it fills and whether it closes the branch. The conclusion remains aligned with CERN-2003-001: the selected arXiv evidence does not connect production to a direct collider survival–capture–growth chain.

Table 4: Equation audit against CERN-2003-001 [1].

Branch step	Equation used here	Evaluation against CERN-2003-001
Production	$\sqrt{\hat{s}} = \sqrt{x_1 x_2 s} > M_{\min} \sim \text{few } M_D$	Aligned with the threshold logic of microscopic black-hole production in TeV-scale gravity models. The corrected variable is the partonic energy $\hat{s}$, not the full proton–proton energy s .
Four-dimensional black-hole size	$R_S = 2GM = M/M_P^2$	Directly matches the CERN-2003-001 four-dimensional argument: ordinary gravity requires masses far above any accelerator energy.
Decay versus accretion	$\Gamma_D \simeq T_{\text{BH}}^4 R_S^2, \quad \Gamma_A \simeq \pi R_S^2 \rho$	Matches CERN’s branch test: growth requires accretion to exceed thermal decay. In the present report this is compressed into the necessary condition $\dot{M}_{\text{net}} > 0$.
Higher-dimensional decay	$T_{\text{BH}} \simeq M_*(M_*/M)^{1/(1+d)}$	Aligned with CERN’s extra-dimensional stability calculation. Low-mass black holes evaporate rapidly; only extremely massive objects survive the bound.
Constructor growth slot	$\dot{M}_{\text{net}} = \rho \sigma_{\text{cap}}(M, v) v - P_{\text{evap}}(M)/c^2$	Rate-balance form of CERN’s decay-versus-accretion test. A collider danger branch must make this balance positive in collider variables.

10 Conclusion

The dangerous LHC black-hole claim fails as a mechanism in the retained equation set. The claim graph shows where black-hole statements occur. The equation graph shows where the physical branch breaks: retained formulas populate adjacent astrophysical mechanisms and growth/capture constraints, plus one production-threshold hook. The missing links are survival after production, stopping or capture in matter, positive growth at microscopic mass and velocity, and evasion of astronomical survival bounds.

The strongest equation support is for adjacent astrophysical mechanisms: accretion, luminosity limits, compact-object masses and survival bounds. A danger argument would have to preserve those equations after changing the mass, velocity, density, capture and lifetime assumptions to the collider case. The retained equations stop before that preservation step.

References

- [1] J.-P. Blaizot, J. Iliopoulos, J. Madsen, G. G. Ross, P. Sonderegger and H.-J. Specht, “Study of potentially dangerous events during heavy-ion collisions at the LHC: Report of the LHC Safety Study Group,” CERN-2003-001 (2003), <https://cds.cern.ch/record/613175/files/CERN-2003-001.pdf>.
- [2] S. B. Giddings and M. L. Mangano, “Astrophysical implications of hypothetical stable TeV-scale black holes,” arXiv:0806.3381 (2008).
- [3] J. Ellis, G. Giudice, M. L. Mangano, I. Tkachev and U. Wiedemann, “Review of the Safety of LHC Collisions,” arXiv:0806.3414 (2008).
- [4] R. Plaga, “On the potential catastrophic risk from metastable quantum-black holes produced at particle colliders,” arXiv:0808.1415 (2008).
- [5] B. Koch, M. Bleicher and H. Stoecker, “Exclusion of black hole disaster scenarios at the LHC,” arXiv:0807.3349 (2008).
- [6] S. B. Giddings and M. L. Mangano, “Comments on claimed risk from metastable black holes,” arXiv:0808.4087 (2008).

- [7] R. Casadio, S. Fabi and B. Harms, “Possibility of catastrophic black hole growth in the warped brane-world scenario at the LHC,” arXiv:0901.2948 (2009).
- [8] S. W. Hawking, “Particle creation by black holes,” *Communications in Mathematical Physics* 43, 199–220 (1975).